

Capstone Project 7

Title: Enhancing Customer Service with Agent-powered Chatbots

Overview and Problem Statement:

This project aims to develop an intelligent multi-agent customer service chatbot system for e-commerce platforms using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic AI workflows. The system will provide fast, personalized, and context-aware customer support across multiple customer touchpoints such as chat interfaces, customer support portals, and web applications.

E-commerce platforms often struggle to handle large volumes of customer queries efficiently. Traditional chatbots frequently fail to provide accurate, contextual, and personalized responses, resulting in poor customer satisfaction, delayed issue resolution, and increased support costs. Businesses also face challenges in managing complex customer scenarios such as order tracking, refunds, returns, product recommendations, and escalation handling.

The proposed system will leverage multi-agent architecture, intelligent reasoning strategies, and dynamic knowledge retrieval to automate customer support workflows and improve overall customer experience.

Dataset:

The project will utilize a combination of publicly available and synthetic datasets:

1. Customer Service Conversations:
 - Source: Kaggle's [Customer Support on Twitter](#) dataset
 - Content: Real customer service interactions from various companies
2. E-commerce Product Catalog:
 - Source: [Amazon product dataset](#) (a subset focused on electronics and home appliances)
 - Content: Product descriptions, specifications, prices, and customer reviews
3. Synthetic E-commerce Queries:
 - Source: Generated using GPT-5 or similar language models
 - Content: Simulated customer queries specific to e-commerce scenarios

Methodology:

1. Data Preparation:



- a. Collect and preprocess customer support conversations and product catalog datasets
 - b. Clean and organize product and customer interaction data
 - c. Generate synthetic customer queries for training and testing scenarios
2. Agentic Architecture Development:
 - a. Design multi-agent workflows for customer query handling
 - b. Create specialized agents for order tracking, returns, escalation handling, and product recommendations
 - c. Implement task decomposition and reasoning strategies
3. RAG Pipeline Integration:
 - a. Build vector database for product and support knowledge retrieval
 - b. Retrieve relevant contextual information during customer interactions
 - c. Integrate semantic search for accurate response generation
4. AI Chatbot Development:
 - a. Develop conversational chatbot using LLMs
 - b. Maintain customer conversation history and context awareness
 - c. Generate personalized and context-aware responses
5. Integration and Automation:
 - a. Build web-based chatbot interface using Streamlit or Gradio
 - b. Integrate APIs for order lookup and customer support workflows
 - c. Automate support workflows and escalation pipelines
6. Evaluation:
 - a. Measure chatbot performance using response accuracy, latency, and customer satisfaction metrics
 - b. Evaluate agent coordination and recommendation quality

Challenges:

1. Task Decomposition: Break complex customer queries into manageable subtasks for specialized agents
2. Multi-Agent Coordination: Ensure efficient collaboration between multiple AI agents
3. Context Retention: Maintain customer conversation history and contextual understanding during interactions
4. Product Knowledge Retrieval: Efficiently manage and retrieve product information using RAG pipelines
5. Scalability: Support high-volume customer interactions with low response latency
6. Personalization: Generate personalized recommendations and responses based on customer history
7. Hallucination and Accuracy: Prevent inaccurate or misleading AI-generated responses

References:

1. Lewis, Patrick, et al. 2020. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." arXiv preprint arXiv:2005.11401.
2. Henderson, Matthew, et al. 2019. "A repository of conversational datasets." arXiv preprint arXiv:1904.06472.
3. Serban, Iulian Vlad, et al. 2017. "A deep reinforcement learning chatbot." arXiv preprint arXiv:1709.02349.